



CSE0613409

Machine Vision and Motion Analysis

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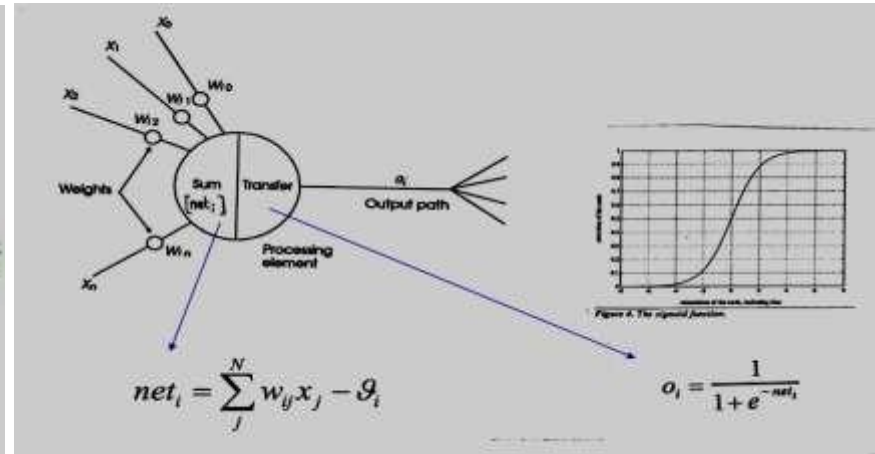
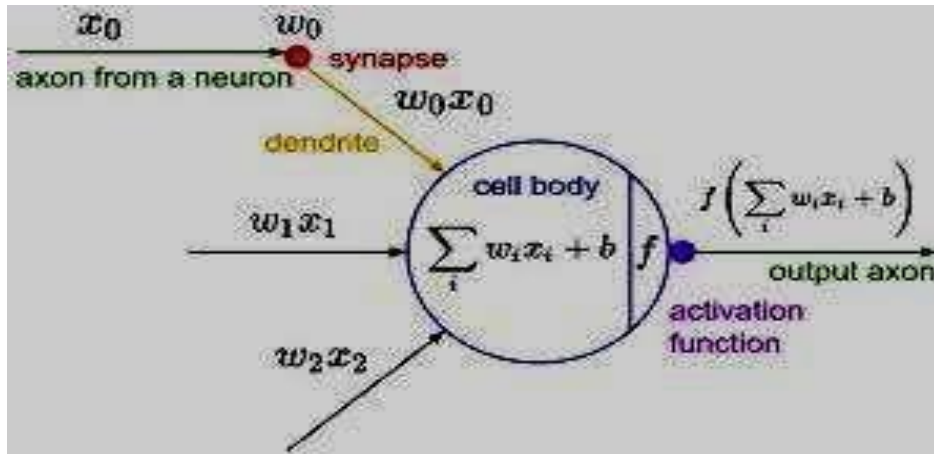
Why Deep Learning is becoming so popular?

- Data Growth (i.e. Software, Apps, Social media)
- Hardware advancements
 - Specialized Hardware (GPU, TPU)
 - Computational in parallel
- Python and open source (Eco system)
- Cloud (i.e. rent a server) and AI Boom



Neuron

- A neuron compute the *weighted average of its input*, and *this sum is passed through a nonlinear function*, often called **activation function**, such as the sigmoid



The output of the neuron can then be sent as input to the neurons of another layer, which could repeat the same computation (weighted sum of the input and transformation with activation function)

Neuron

age	have_insurance
22	0
25	0
47	1
52	0
46	1
56	1
55	0
60	1
62	1
61	1
18	0
28	0
27	0
29	0
49	1

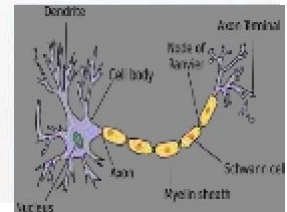


Neuron

age	have_insurance
22	0
25	0
47	1
52	0
46	1
56	1
55	0
60	1
62	1
61	1
18	0
28	0
27	0
29	0
49	1

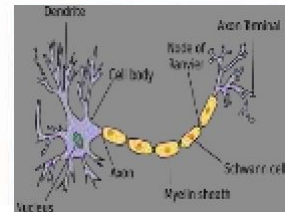
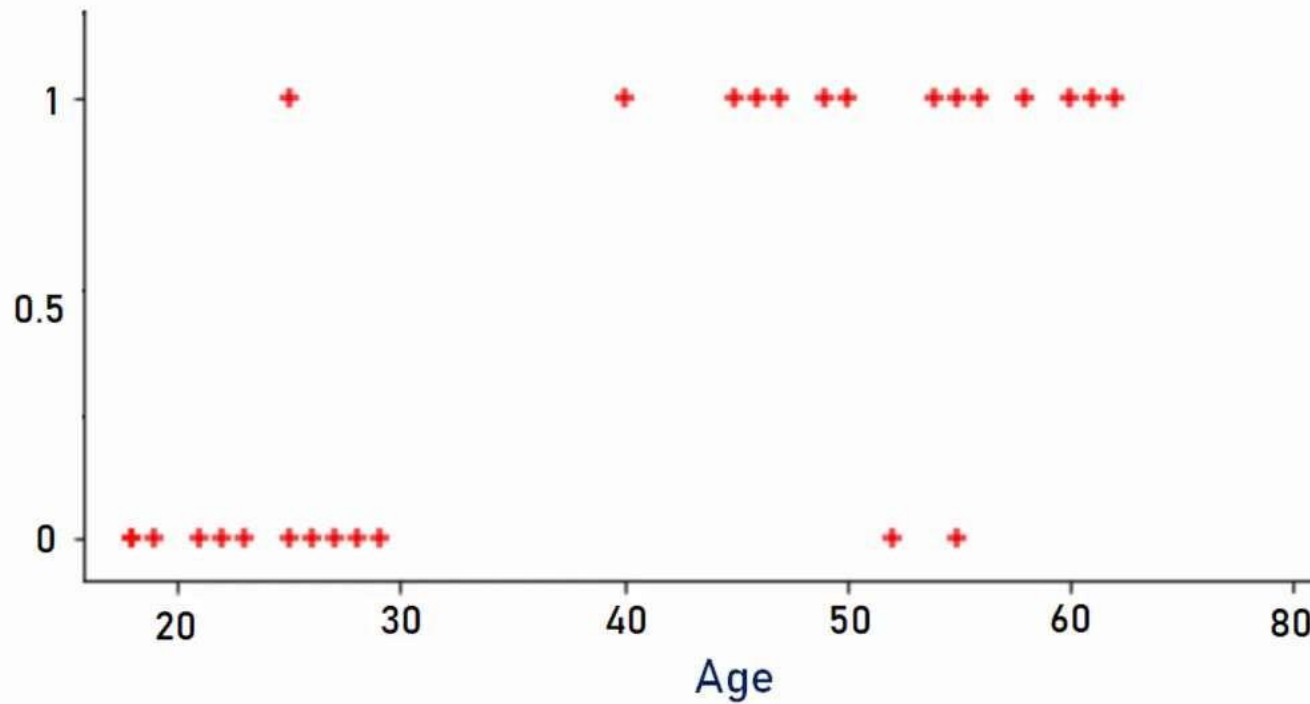
Binary Classification

Given an age of a person, come up with a **function** that can predict if person will buy insurance or not



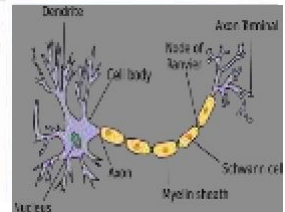
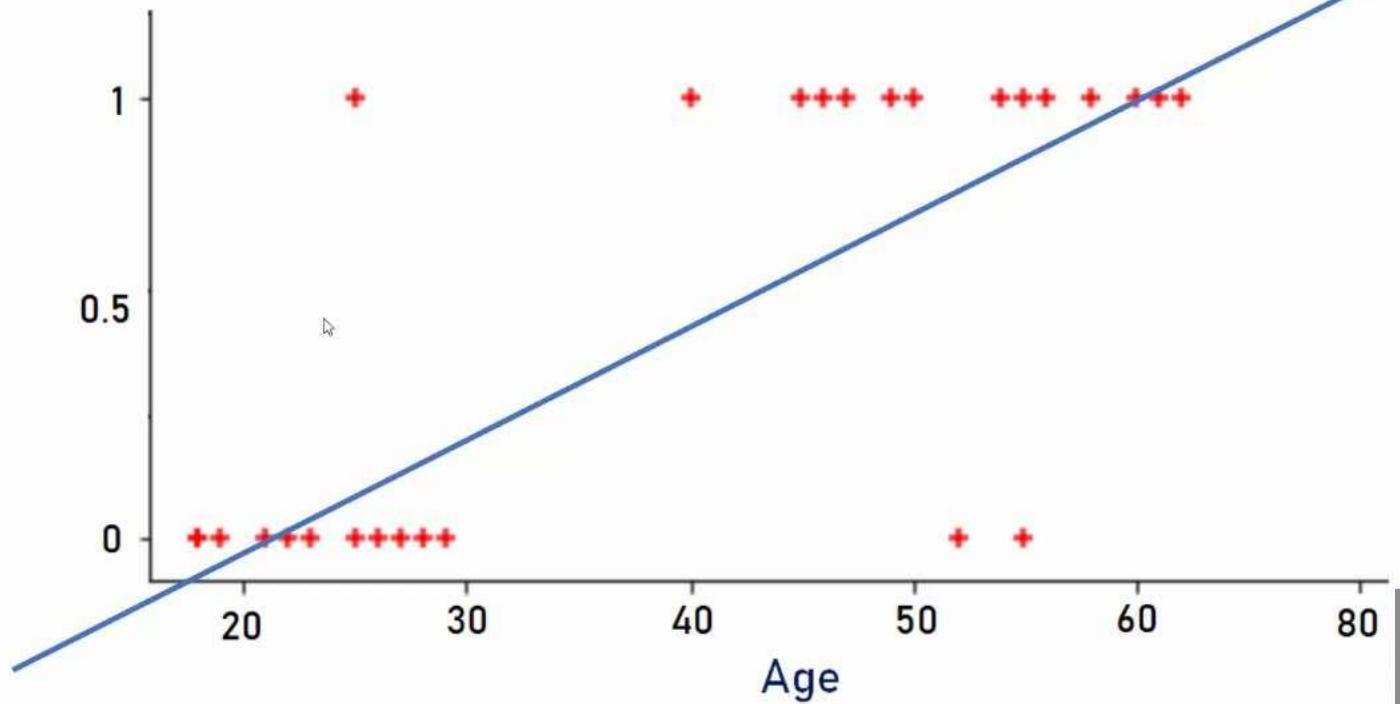
Neuron

Have
Insurance?



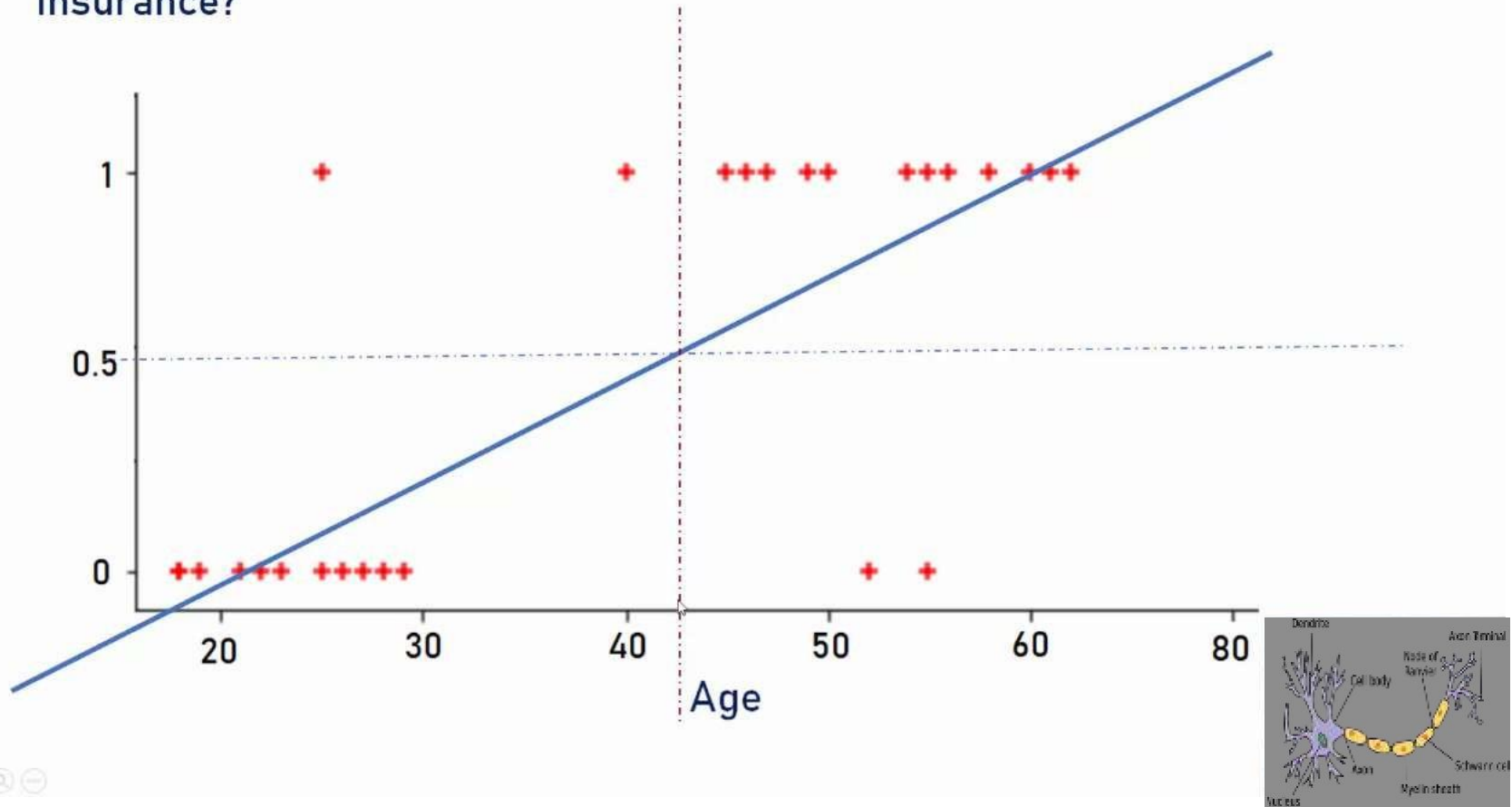
Neuron

Have Insurance?

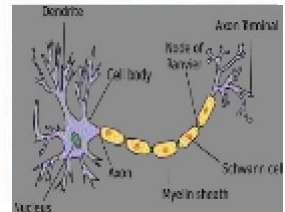
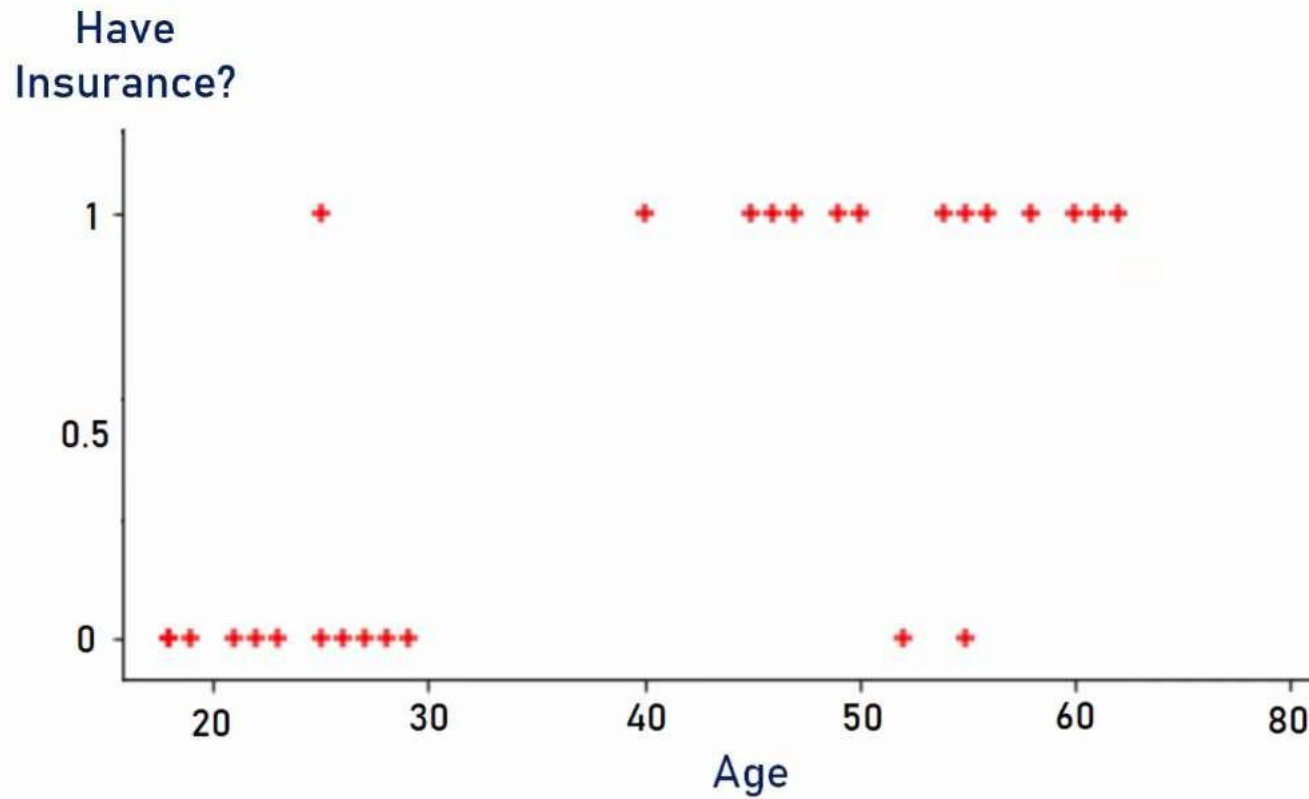


Neuron

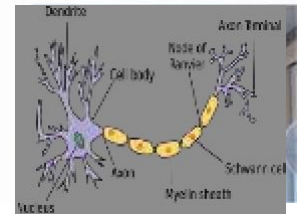
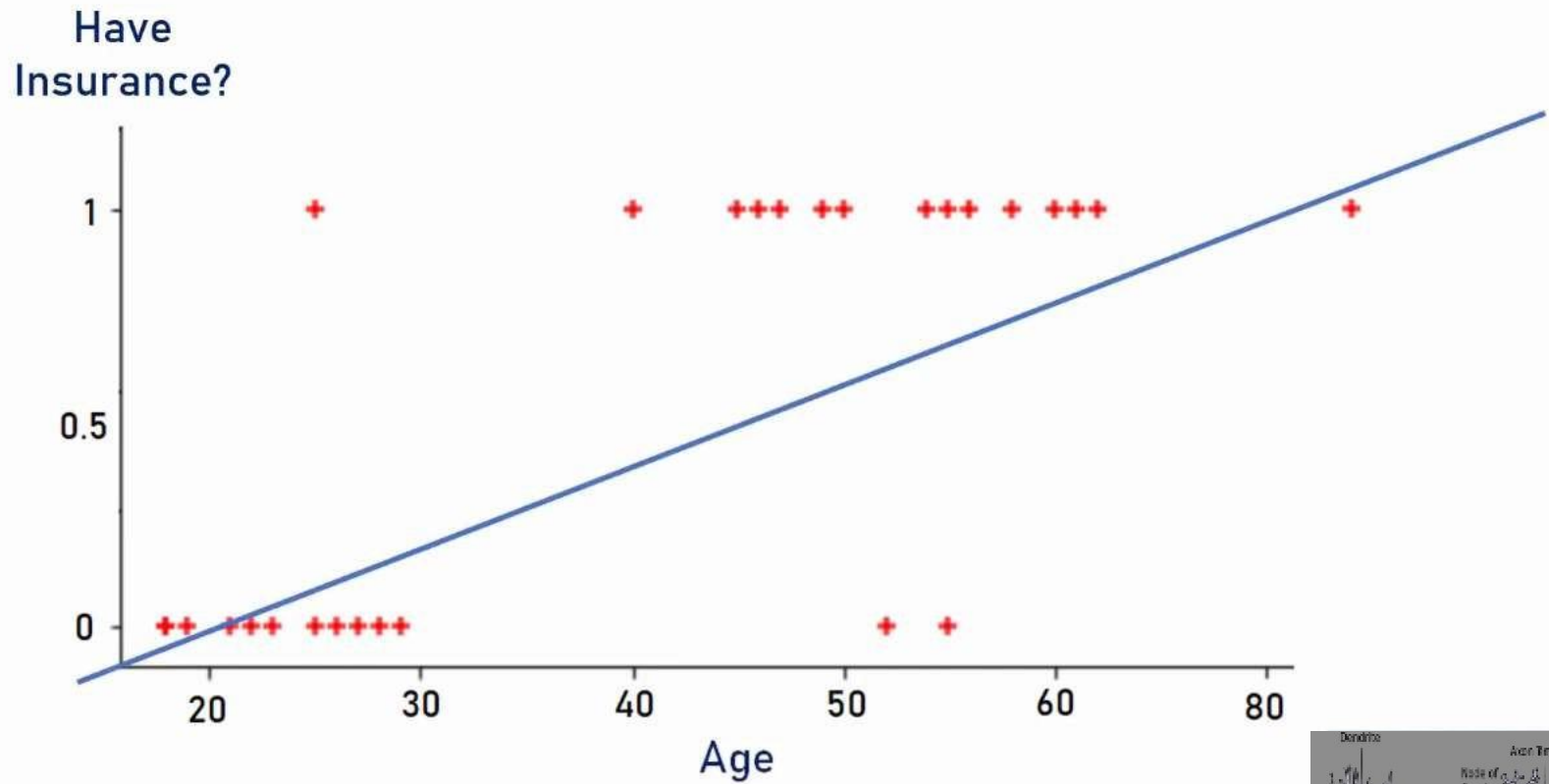
Have
Insurance?



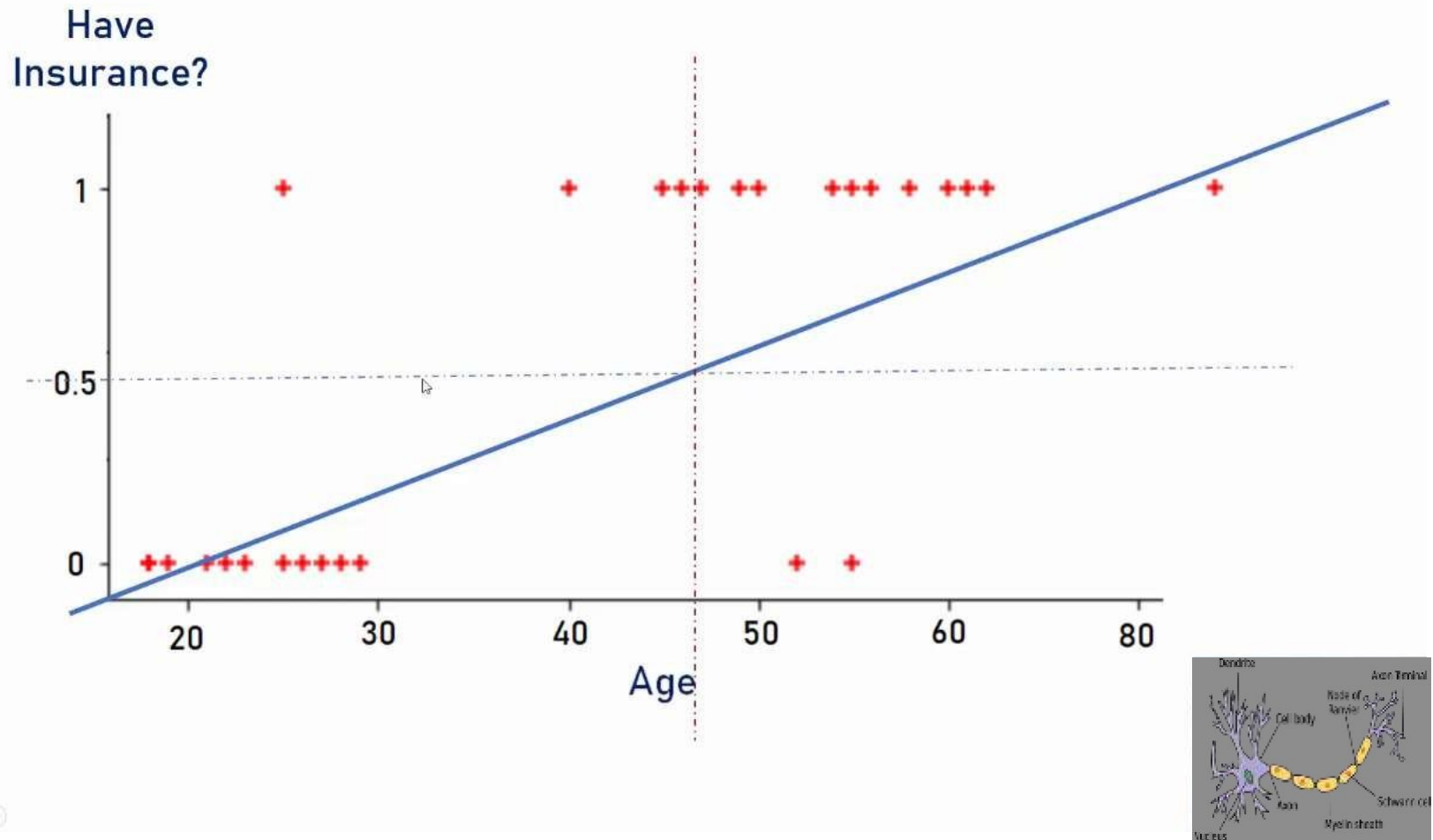
Neuron



Neuron

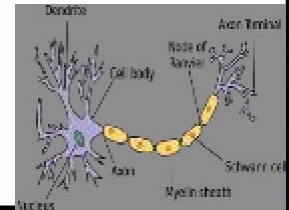
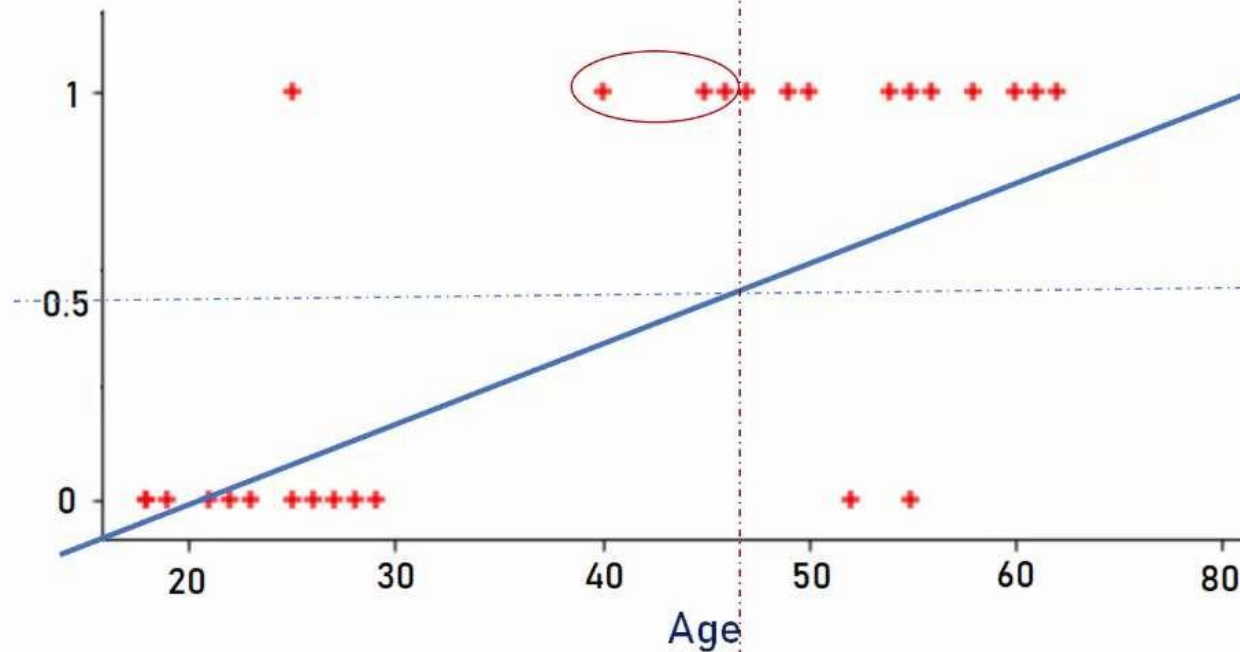


Neuron



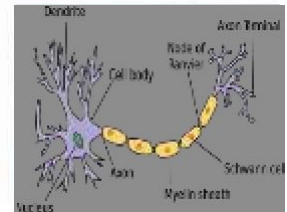
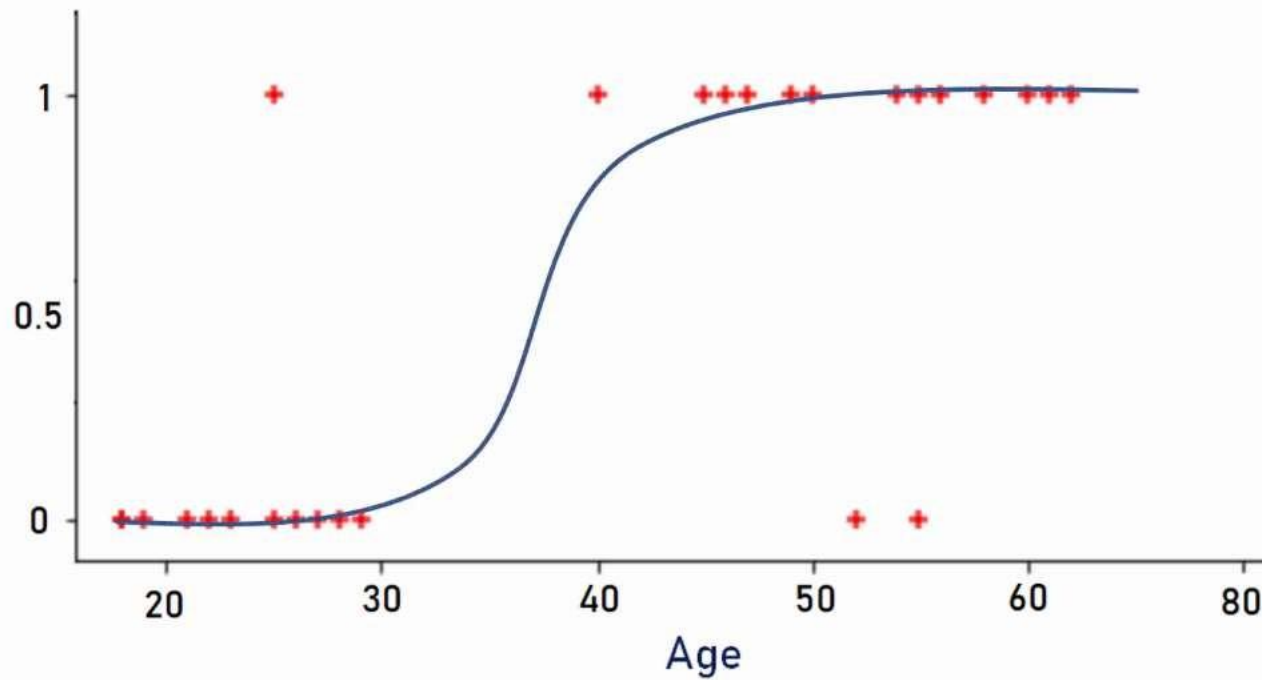
Neuron

Have Insurance?



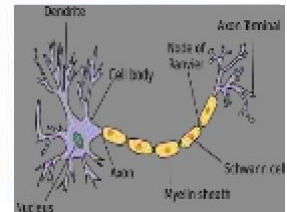
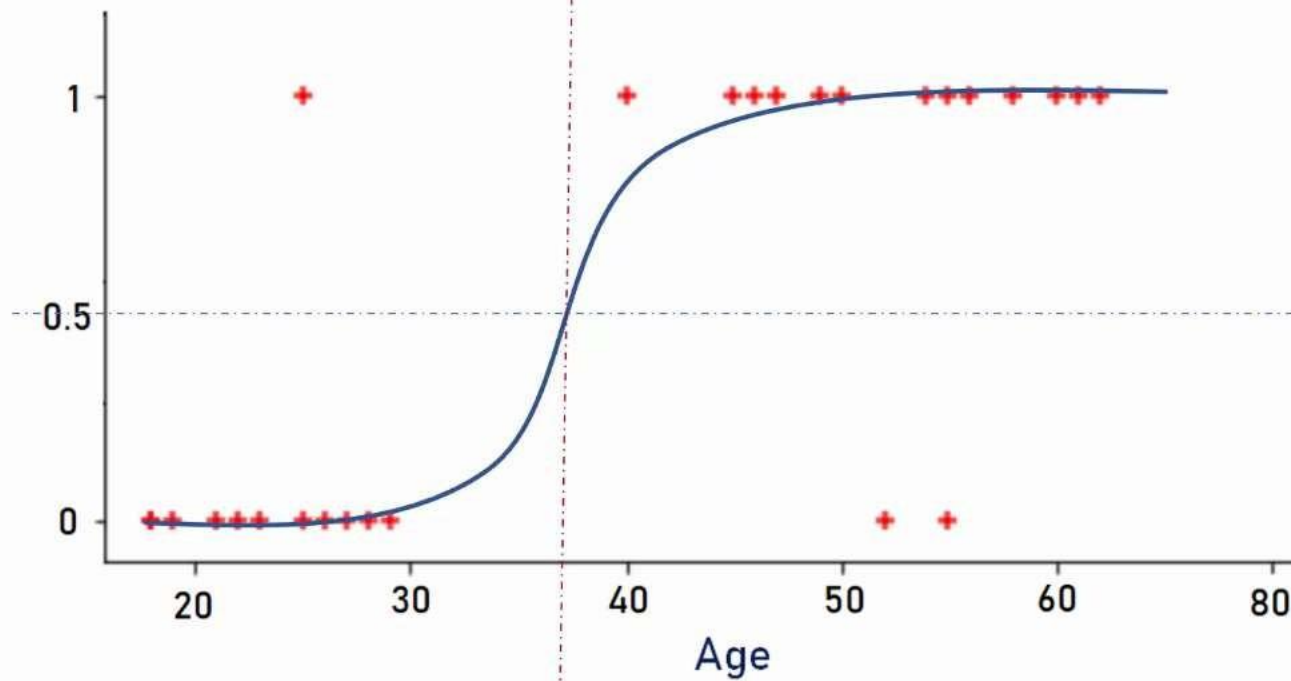
Neuron

Have
Insurance?

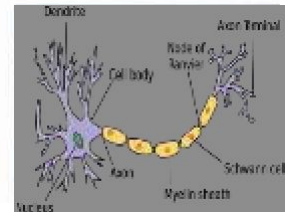
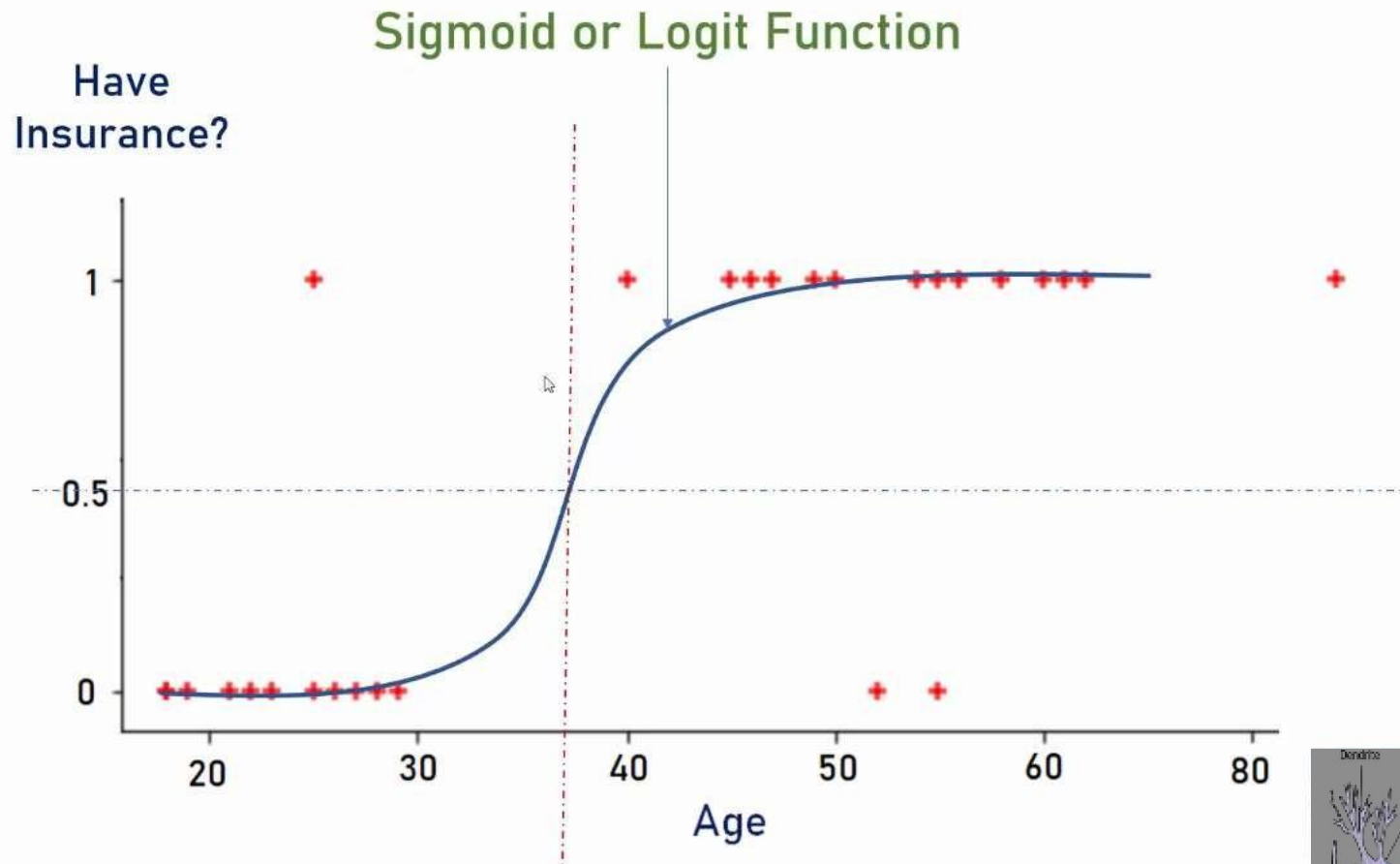


Neuron

Have Insurance?



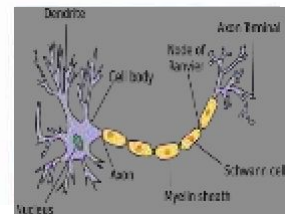
Neuron



Neuron

$$\textit{sigmoid}(z) = \frac{1}{1 + e^{-z}}$$

e = Euler's number ~ 2.71828



Neuron

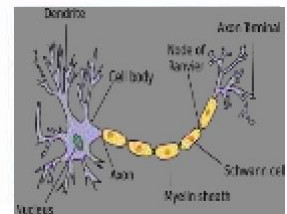
$$\text{sigmoid}(z) = \frac{1}{1 + e^{-z}}$$

e = Euler's number ~ 2.71828

$$\text{sigmoid}(200) = \frac{1}{1 + 2.71^{-200}} = \text{almost close to } 1$$

$$\text{sigmoid}(-200) = \frac{1}{1 + 2.71^{200}} = \text{almost close to } 0$$

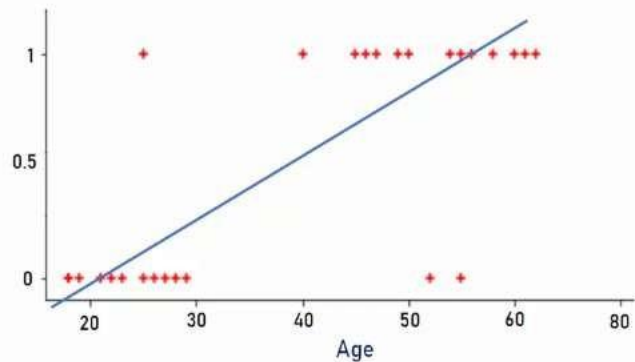
Sigmoid function converts input into range 0 to 1



Neuron

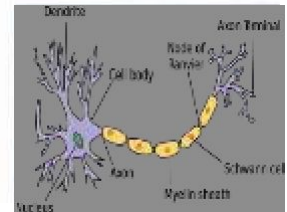
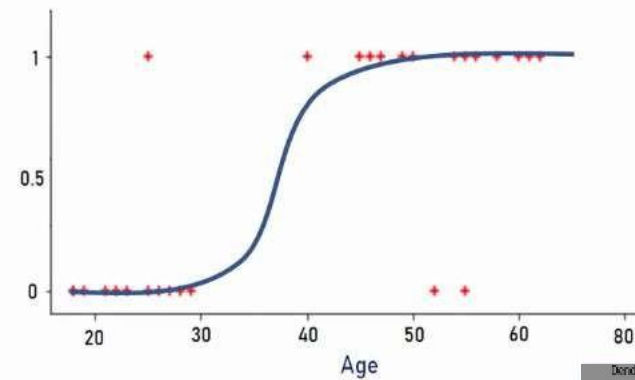
Step 1

$$y = m * x + b$$



Step 2

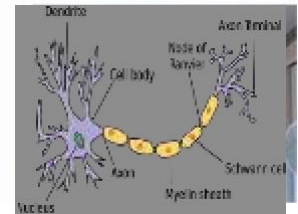
$$z = \frac{1}{1 + e^{-y}}$$



Neuron

$$y = 0.042 * x - 1.53$$

Age



Neuron

1_log 1_log 1_log (845) 1_log 1_log Install Home python Home Unbitl Home Unbitl Home Unbitl 7_log Anim Mane (877) (877) py x + - □ ×

github.com/codebasics/py/blob/master/ML/7_logistic_reg/7_logistic_regression.ipynb

24	50
25	54

model.coef_ indicates value of m in $y=m*x + b$ equation

In [67]: `model.coef_`

Out[67]: `array([[0.04150133]])`

model.intercept_ indicates value of b in $y=m*x + b$ equation

In [68]: `model.intercept_`

Out[68]: `array([-1.52726963])`

Lets defined sigmoid function now and do the math with hand

In [43]: `import math`
`def sigmoid(x):`
 `return 1 / (1 + math.exp(-x))`

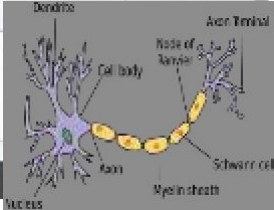
In [75]: `def prediction_function(age):`
 `z = 0.042 * age - 1.53 # 0.04150133 ~ 0.042 and -1.52726963 ~ -1.53`
 `y = sigmoid(z)`
 `return y`

In [76]: `age = 35`
`prediction_function(age)`

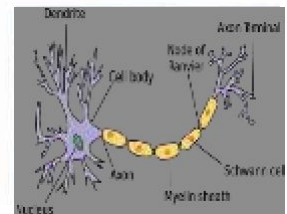
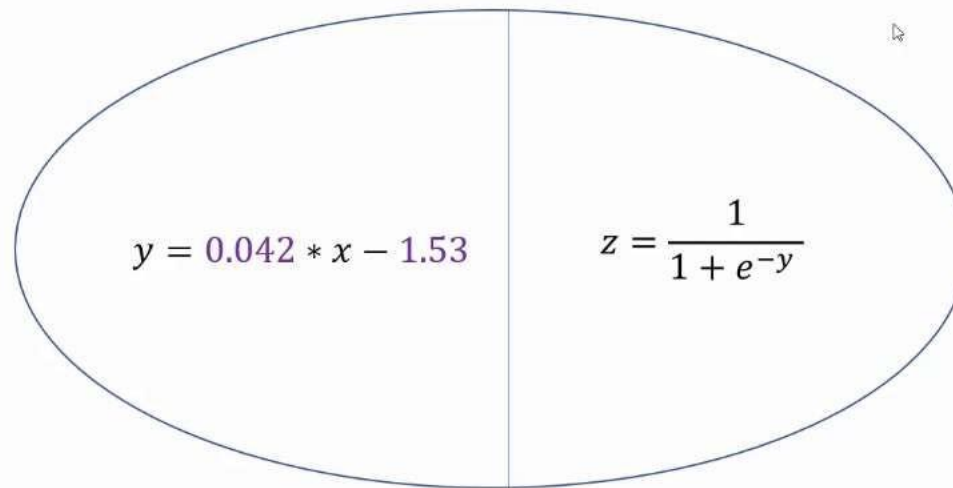
Out[76]: `0.4850044983805899`

0.485 is less than 0.5 which means person with 35 age will *not* buy insurance

In [77]: `age = 42`



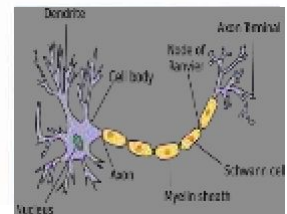
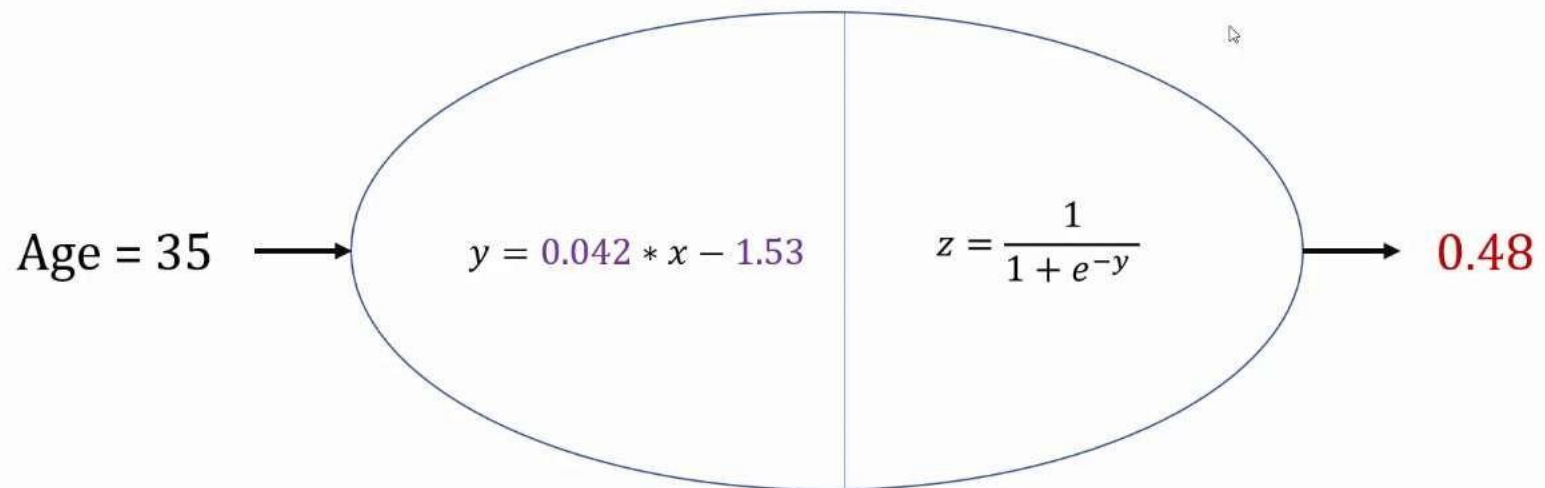
Neuron



Neuron

value < 0.5 = person will not buy insurance

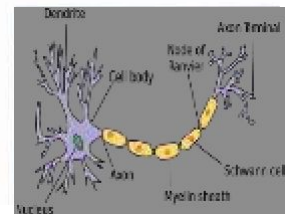
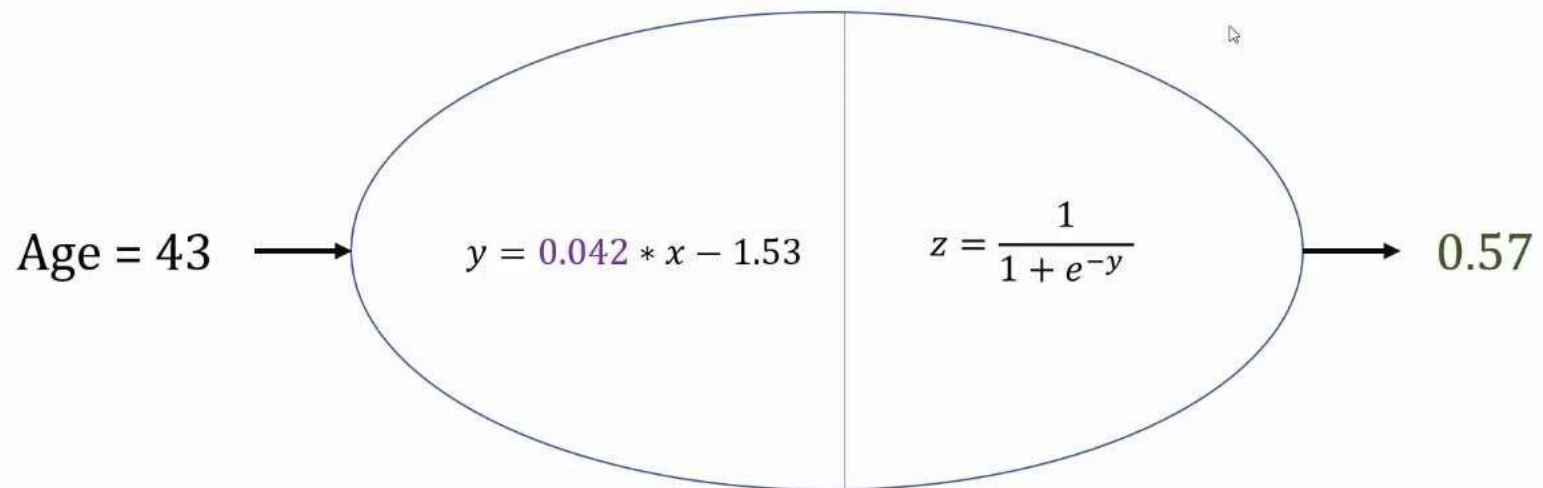
value ≥ 0.5 = person **will** buy insurance



Neuron

value < 0.5 = person will not buy insurance

value ≥ 0.5 = person **will** buy insurance



Neuron

$$y = 0.042 * x - 1.53$$

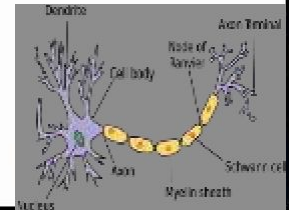
Age

$$y = 0.042 * x1 + 0.008 * x2 + 0.2 * x3 - 1.53$$

Age Income Education

$$y = w1 * x1 + w2 * x2 + w3 * x3 + b$$

$$y = \sum_{i=0}^n w^i x^i + b$$



Neuron

